

COSC 594

Policy Optimization (III)

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Outline

- RECAP
- NATURAL POLICY GRADIENT METHODS
- GLOBAL CONVERGENCE ANALYSIS

Outline

■ RECAP

Natural policy gradient

$$F_{\rho}^{\dagger}(\theta) \nabla_{\theta} V^{\pi_{\theta}}(\rho)$$

■ FISHER INFORMATION MATRIX

$$F_{\rho}(\theta) \quad := \quad \mathbb{E}_{s \sim d_{\rho}^{\pi_{\theta}}(\cdot)} \mathbb{E}_{a \sim \pi_{\theta}(\cdot | s)} \left[\nabla_{\theta} \log \pi_{\theta}(a | s) (\nabla_{\theta} \log \pi_{\theta}(a | s))^{\top} \right]$$

$\nabla_{\theta} \log \pi_{\theta}(a | s)$ – Fisher score

$d_{\rho}^{\pi_{\theta}}(\cdot)$ – stationary state distribution

Natural policy gradient methods

Policy: $\pi_{\theta}(\cdot | s): S \rightarrow \Delta_A$

Parameter: $\theta \in \mathbb{R}^d$

■ NONCONVEX PARAMETER OPTIMIZATION

$$\underset{\theta}{\text{maximize}} \quad V^{\pi_{\theta}}(\rho)$$

■ NATURAL GRADIENT ASCENT

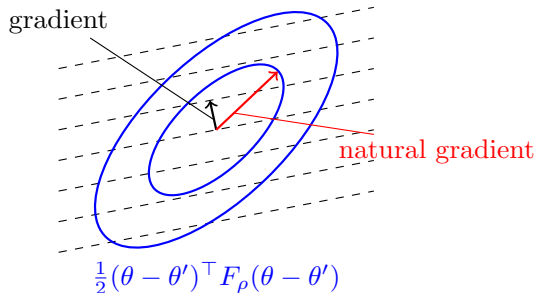
$$\theta^+ \leftarrow \theta + \eta F_{\rho}^{\dagger}(\theta) \nabla_{\theta} V^{\pi_{\theta}}(\rho)$$

- ★ Search the decision in **a parameter space**
- ★ **Costly inverse** of Fisher information matrix

Natural policy gradient

Natural policy gradient: $F_{\rho}(\theta)^{\dagger} \nabla_{\theta} V^{\pi_{\theta}}(\rho)$

Fisher information: $F_{\rho}(\theta) = \mathbb{E} \left[\nabla_{\theta} \log \pi_{\theta} (\nabla_{\theta} \log \pi_{\theta})^{\top} \right]$



Steepest ascent in Fisher information distance

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Natural policy gradient: multiplicative weights update

$$\pi_{\theta}^{+}(a | s) = \pi_{\theta}(a | s) \frac{\exp\left(\frac{\eta}{1-\gamma} A^{\pi_{\theta}}(s, a)\right)}{Z^{\pi_{\theta}}(s)}$$

$Z^{\pi_{\theta}}(s) := \sum_{a \in A} \pi_{\theta}(a | s) \exp\left(\frac{\eta}{1-\gamma} A^{\pi_{\theta}}(s, a)\right)$ – normalization constant

$A^{\pi_{\theta}}(s, a) = Q^{\pi_{\theta}}(s, a) - V^{\pi_{\theta}}(s)$ – advantage value function

■ NATURAL POLICY GRADIENT

$$\begin{aligned}(1 - \gamma) F_{\rho}^{\dagger}(\theta) \nabla_{\theta} V^{\pi_{\theta}}(\rho) &= F_{\rho}^{\dagger}(\theta) \mathbb{E}_{s \sim d_{\rho}^{\pi_{\theta}}(\cdot), a \sim \pi_{\theta}(\cdot | s)} [A^{\pi_{\theta}}(s, a) \nabla_{\theta} \log \pi_{\theta}(a | s)] \\ &= \operatorname{argmin}_w \mathbb{E}_{s \sim d_{\rho}^{\pi_{\theta}}(\cdot), a \sim \pi_{\theta}(\cdot | s)} \left[\left(A^{\pi_{\theta}}(s, a) - w^{\top} \nabla_{\theta} \log \pi_{\theta}(a | s) \right)^2 \right]\end{aligned}$$

■ SOFTMAX POLICY STRUCTURE

$$\left(w^\top \nabla_\theta \log \pi_\theta(a | s)\right)_{s,a} = w_{s,a} - \sum_{a'} w_{s,a'} \pi_\theta(a' | s) := w_{s,a} - \bar{w}_s.$$

$\Downarrow$

$$w_{s,a}^\star = A^{\pi_\theta}(s, a) + \bar{w}_s^\star$$

$\Downarrow$

$$(F_\rho^\dagger(\theta) \nabla_\theta V^{\pi_\theta}(\rho))_{s,a} = \frac{1}{1-\gamma} w_{s,a}^\star = \frac{1}{1-\gamma} A^{\pi_\theta}(s, a) + \frac{1}{1-\gamma} \bar{w}_s^\star$$

$\Downarrow$

$$\pi_{\theta+}(a | s) = \frac{e^{\theta_{s,a}^+}}{\sum_{a'} e^{\theta_{s,a'}^+}} = \frac{e^{\theta_{s,a} + \frac{\eta}{1-\gamma} A^{\pi_\theta}(s,a)}}{\sum_{a'} e^{\theta_{s,a'} + \frac{\eta}{1-\gamma} A^{\pi_\theta}(s,a')}}.$$

★ Soft policy iteration

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Global convergence

Natural gradient ascent : $\theta^+ = \theta + \eta F_\rho^\dagger(\theta) \nabla_\theta V^{\pi_\theta}(\rho)$

Multiplicative weights update : $\pi_\theta^+(a | s) = \pi_\theta(a | s) \frac{\exp\left(\frac{\eta}{1-\gamma} A^{\pi_\theta}(s, a)\right)}{Z^{\pi_\theta}(s)}$

Incremental update: $\theta_{s,a}^+ = \theta_{s,a} + \frac{\eta}{1-\gamma} A^{\pi_\theta}(s, a)$

■ SUBLINEAR RATE

$$V^{\pi^\star}(\rho) - V^{\pi^{(T)}}(\rho) \leq \frac{\log |A|}{\eta T} + \frac{1}{(1-\gamma)^2 T}$$

★ No state-dependence and logarithmic action-dependence

■ PERFORMANCE DIFFERENCE

$$\begin{aligned} V^\pi(s) - V^{\pi'}(s) &= \mathbb{E}_{\tau \sim \Pr(\cdot | \pi)} \left[\sum_{t=0}^{\infty} \gamma^t r(s_t, a_t) \right] - V^{\pi'}(s) \\ &= \mathbb{E}_{\tau \sim \Pr(\cdot | \pi)} \left[\sum_{t=0}^{\infty} \gamma^t \left(r(s_t, a_t) + V^{\pi'}(s_t) - V^{\pi'}(s_t) \right) \right] - V^{\pi'}(s) \\ &= \mathbb{E}_{\tau \sim \Pr(\cdot | \pi)} \left[\sum_{t=0}^{\infty} \gamma^t \left(r(s_t, a_t) + \gamma V^{\pi'}(s_{t+1}) - V^{\pi'}(s_t) \right) \right] \\ &= \mathbb{E}_{\tau \sim \Pr(\cdot | \pi)} \left[\sum_{t=0}^{\infty} \gamma^t \left(Q^{\pi'}(s_t, a_t) - V^{\pi'}(s_t) \right) \right] \\ &= \mathbb{E}_{\tau \sim \Pr(\cdot | \pi)} \left[\sum_{t=0}^{\infty} \gamma^t A^{\pi'}(s_t, a_t) \right] \\ &= \frac{1}{1 - \gamma} \mathbb{E}_{s' \sim d_s^\pi(\cdot), a' \sim \pi(\cdot | s)} \left[A^{\pi'}(s', a') \right] \end{aligned}$$

■ POLICY IMPROVEMENT PER STEP

$$\begin{aligned} V^{(t+1)}(\mu) - V^{(t)}(\mu) &= \frac{1}{1-\gamma} \mathbb{E}_{s \sim d_{\mu}^{(t+1)}(\cdot), a \sim \pi^{(t+1)}(\cdot | s)} \left[A^{(t)}(s, a) \right] \\ &= \frac{1}{\eta} \mathbb{E}_{s \sim d_{\mu}^{(t+1)}(\cdot)} \left[\sum_a \pi^{(t+1)}(a | s) \log \frac{\pi^{(t+1)}(a | s) Z^{(t)}(s)}{\pi^{(t)}(a | s)} \right] \\ &= \frac{1}{\eta} \mathbb{E}_{s \sim d_{\mu}^{(t+1)}(\cdot)} \left[D_{\text{KL}}(\pi^{(t+1)}(\cdot | s) \| \pi^{(t)}(\cdot | s)) + \log Z^{(t)}(s) \right] \\ &\geq \frac{1}{\eta} \mathbb{E}_{s \sim d_{\mu}^{(t+1)}(\cdot)} \left[\log Z^{(t)}(s) \right] \\ &\geq \frac{1-\gamma}{\eta} \mathbb{E}_{s \sim \mu(\cdot)} \left[\log Z^{(t)}(s) \right] \\ &\geq 0 \end{aligned}$$

$d_{\mu}^{(t+1)} \geq (1-\gamma)\mu$ – componentwise lower bound

★ Monotonic policy improvement

■ POLICY OPTIMALITY PER STEP

$$\begin{aligned} V^*(\mu) - V^{(t)}(\mu) &= \frac{1}{1 - \gamma} \mathbb{E}_{s \sim d_\mu^*(\cdot), a \sim \pi^*(\cdot | s)} \left[A^{(t)}(s, a) \right] \\ &= \frac{1}{\eta} \mathbb{E}_{s \sim d_\mu^*(\cdot)} \left[\sum_a \pi^*(a | s) \log \frac{\pi^{(t+1)}(a | s) Z^{(t)}(s)}{\pi^{(t)}(a | s)} \right] \\ &= \frac{1}{\eta} \mathbb{E}_{s \sim d_\mu^*(\cdot)} \left[D_{\text{KL}}(\pi^*(\cdot | s) \| \pi^{(t)}(\cdot | s)) - D_{\text{KL}}(\pi^*(\cdot | s) \| \pi^{(t+1)}(\cdot | s)) \right] \\ &\quad + \frac{1}{\eta} \mathbb{E}_{s \sim d_\mu^*(\cdot)} \left[\log Z^{(t)}(s) \right] \end{aligned}$$

$\Downarrow$

$$\begin{aligned} \frac{1}{T} \sum_{t=0}^{T-1} \left(V^*(\rho) - V^{(t)}(\rho) \right) &\leq \frac{1}{\eta T} \sum_{t=0}^{T-1} \mathbb{E}_{d_\rho^*} \left[D_{\text{KL}}(\pi_s^* \| \pi_s^{(t)}) - D_{\text{KL}}(\pi_s^* \| \pi_s^{(t+1)}) \right] \\ &\quad + \frac{1}{\eta T} \sum_{t=0}^{T-1} \mathbb{E}_{d_\rho^*} \left[\log Z^{(t)}(s) \right] \end{aligned}$$

■ MONOTONICITY + POLICY OPTIMALITY

$$\begin{aligned} V^*(\rho) - V^{(T-1)}(\rho) &\leq \frac{1}{T} \sum_{t=0}^{T-1} \left(V^*(\rho) - V^{(t)}(\rho) \right) \\ &\leq \frac{1}{\eta T} \sum_{t=0}^{T-1} \mathbb{E}_{d_\rho^*} \left[D_{\text{KL}}(\pi_s^* \parallel \pi_s^{(t)}) - D_{\text{KL}}(\pi_s^* \parallel \pi_s^{(t+1)}) \right] \\ &\quad + \frac{1}{\eta T} \sum_{t=0}^{T-1} \mathbb{E}_{d_\rho^*} \left[\log Z^{(t)}(s) \right] \\ &\leq \frac{1}{\eta T} \mathbb{E}_{d_\rho^*} \left[D_{\text{KL}}(\pi_s^* \parallel \pi_s^{(0)}) \right] + \frac{1}{(1-\gamma)T} \sum_{t=0}^{T-1} \left(V^{(t+1)}(d_\rho^*) - V^{(t)}(d_\rho^*) \right) \\ &\leq \frac{1}{\eta T} \mathbb{E}_{d_\rho^*} \left[D_{\text{KL}}(\pi_s^* \parallel \pi_s^{(0)}) \right] + \frac{1}{(1-\gamma)T} V^{(T)}(d_\rho^*) \\ &\leq \frac{1}{\eta T} \log |A| + \frac{1}{(1-\gamma)^2 T} \end{aligned}$$

Generalization

Natural gradient ascent : $\theta^+ = \theta + \eta F_\rho^\dagger(\theta) \nabla_\theta V^{\pi_\theta}(\rho)$

Multiplicative weights update : $\pi_\theta^+(a | s) = \pi_\theta(a | s) \frac{\exp\left(\frac{\eta}{1-\gamma} A^{\pi_\theta}(s, a)\right)}{Z^{\pi_\theta}(s)}$

Incremental update: $\theta_{s,a}^+ = \theta_{s,a} + \frac{\eta}{1-\gamma} A^{\pi_\theta}(s, a)$

■ POLICY MIRROR DESCENT

$$\pi^+(\cdot | s) \leftarrow \operatorname{argmax}_{\pi'} \langle \mathbf{A}^\pi(s, \cdot), \pi'(\cdot | s) - \pi(\cdot | s) \rangle - \frac{1}{\eta} D_{\text{KL}}(\pi'(\cdot | s) \| \pi(\cdot | s))$$

★ Mirror descent over a function space

Summary

■ RECAP

Natural policy gradient

■ NATURAL POLICY GRADIENT METHODS

Multiplicative weights update

■ GLOBAL CONVERGENCE ANALYSIS

Global last-iterate convergence